# ASSIGNMENT: SE – OVERVIEW OF IT INDUSTRY

## Session 1 — IT Industry Basics

### 1. Name 5 IT job roles and what they do

| IT Job Role | What they do |
|---|---|
| 1. Software Developer | Develops, tests, and maintains software applications. |
| 2. Web Developer | Creates and maintains websites and web applications. |
| 3. Database Administrator (DBA) | Manages databases, including data storage, security, backup, and performance. |
| 4. Software Tester / QA Engineer | Tests software to find bugs and ensures the application works correctly. |
| 5. DevOps Engineer | Manages software deployment, servers, automation, and CI/CD processes. |

### 2. Product Companies vs Service Companies

- **Product Companies**

| Company | Example of Work |
|---|---|
| Microsoft | Develops products such as Windows, Microsoft 365, and Azure. |
| Google | Develops products such as Google Search, YouTube, and Android. |
| Adobe | Develops software products such as Photoshop and Acrobat. |

- **Service Companies**

| Company | Example of Work |
|---|---|
| TCS | Provides IT consulting and software development services to clients. |
| Infosys | Provides IT consulting, software development, and digital services. |
| Wipro | Provides IT services, consulting, and technology solutions to businesses. |

- **Easy difference:**
  - **Product company** → Creates and sells its own software/product.
  - **Service company** → Creates or manages software for clients/customers.

### 3. Full forms and one-line use

| Term | Full Form | One-line Use |
|---|---|---|
| GitHub | GitHub is not an acronym; it is a platform name | Used to host, manage, and collaborate on source code using Git. |
| Slack | Slack is a product/platform name, not a standard full form | Used for team communication and collaboration. |
| Jira | Jira is a product name, not an acronym | Used to track tasks, bugs, and software development projects. |
| IDE | Integrated Development Environment | Used to write, run, debug, and manage code. |
| API | Application Programming Interface | Allows different software applications to communicate with each other. |

### 4. Five programming languages and popular uses

| Programming Language | Popular Use Case |
|---|---|
| Java | Enterprise applications and backend development. |
| JavaScript | Web development and interactive websites. |
| Python | AI, Machine Learning, data science, and backend development. |
| C | System programming and embedded systems. |
| C++ | Game development and high-performance applications. |

### 5. Developer vs Tester

- **Developer:** Writes and maintains the code to build software and features.
- **Tester:** Tests the software to find bugs and checks whether it works according to requirements.

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## Session 2 — Software Architecture & Applications

### 1. Three main layers of software architecture

- **Presentation Layer** – The user interface where users interact with the application.
- **Business Logic Layer** – Contains the application rules, logic, and processing.
- **Data Layer** – Handles storing, retrieving, and managing data in the database.
- **Simple flow:**

  User → Presentation Layer → Business Logic Layer → Data Layer → Database

### 2. Client and Server

- **Client:** A device or application that sends a request for a service or data.
- **Server:** A computer or system that receives the request, processes it, and sends a response.
- **Real-life example:**

  When you open YouTube on your phone, your phone is the client, and YouTube's server sends videos and other data back to your phone.

### 3. Desktop Apps vs Web Apps vs Mobile Apps

| Type | Difference 1 | Difference 2 |
|---|---|---|
| Desktop App | Installed directly on a computer/laptop. | Usually designed for Windows, macOS, or Linux. |
| Web App | Runs inside a web browser. | Usually does not require installation; accessed through a URL. |
| Mobile App | Designed for smartphones/tablets. | Usually installed from an app store such as Google Play or App Store. |

- **Examples:**
  - Desktop → Microsoft Word
  - Web → Gmail in a browser
  - Mobile → WhatsApp mobile app

### 4. Three databases and their type

| Database | Type |
|---|---|
| MySQL | SQL |
| PostgreSQL | SQL |
| MongoDB | NoSQL |

- **Easy way to remember:**
  - **SQL** → Data is generally stored in tables (rows and columns).
  - **NoSQL** → Data can be stored in formats such as documents, key-value pairs, or graphs.

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## Session 3 — Networking Basics

### 1. Full forms and one-line use

| Protocol | Full Form | One-line Use |
|---|---|---|
| HTTP | HyperText Transfer Protocol | Used to transfer web pages and data between a web browser and a web server. |
| HTTPS | HyperText Transfer Protocol Secure | Used to securely transfer data between a browser and server using encryption. |
| FTP | File Transfer Protocol | Used to upload and download files between computers over a network. |
| SMTP | Simple Mail Transfer Protocol | Used to send emails from a mail client to a mail server or between mail servers. |

### 2. Difference between IPv4 and IPv6

- IPv4 uses a 32-bit address, such as 192.168.1.1, and provides about 4.3 billion addresses.
- IPv6 uses a 128-bit address, such as 2001:db8::1, and provides a much larger number of addresses.

### 3. nslookup google.com

- Run this command in Command Prompt (CMD):

  nslookup google.com

- The IP address can vary depending on your location, DNS server, and time. You may get output similar to:

  Name: google.com

  Addresses: 142.250.195.14

**Note:** Write the IP address shown on your own system, because google.com can return different IP addresses.

### 4. HTTP Status Codes

| Code | Meaning | Explanation |
|---|---|---|
| 200 | OK | Request was successful. |
| 301 | Moved Permanently | The requested resource has permanently moved to a new URL. |
| 404 | Not Found | The requested page/resource does not exist. |
| 500 | Internal Server Error | An unexpected error occurred on the server. |
| 403 | Forbidden | The server understood the request but refuses to allow access. |

## 5. DNS Hierarchy using www.google.com

- DNS (Domain Name System) converts domain names into IP addresses.

  For www.google.com:

  Root (.)

  ↓

  TLD (.com)

  ↓

  Domain (google)

  ↓

  Subdomain (www)

- **Root (.)** → The top level of the DNS hierarchy.
- **TLD (.com)** → Top-Level Domain; .com is the TLD here.
- **Domain (google)** → The main registered domain name.
- **Subdomain (www)** → A subdomain used to access Google's web service.
- **Simple example:** www.google.com = www (subdomain) + google (domain) + .com (TLD) + root (.).

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## Session 4 — SDLC, Agile, Waterfall & Scrum

### 1. Six phases of SDLC in order

- **SDLC** = Software Development Life Cycle
  1. **Planning** – Identify the project goals, requirements, resources, and timeline.
  2. **Requirement Analysis** – Collect and analyze what the customer needs from the software.
  3. **Design** – Create the software architecture, database design, UI design, and technical structure.
  4. **Development / Implementation** – Developers write the code and build the software.
  5. **Testing** – Test the software to find and fix bugs and ensure it meets requirements.
  6. **Deployment & Maintenance** – Release the software to users and maintain or update it after release.
- **Flow:**

  Planning → Requirement Analysis → Design → Development → Testing → Deployment & Maintenance

### 2. Three differences between Agile and Waterfall

| Agile | Waterfall |
|---|---|
| Development is iterative and incremental. | Development follows a fixed sequential process. |
| Requirements can change during development. | Requirements are generally defined early and changes are difficult. |
| Testing happens continuously during development. | Testing mainly happens after development is completed. |

**Easy example:**

- **Agile =** Build → Test → Get feedback → Improve → Repeat
- **Waterfall =** Requirements → Design → Development → Testing → Deployment

### 3. Three main Scrum roles

- **Product Owner –** Manages the product requirements and prioritizes the Product Backlog.
- **Scrum Master –** Helps the Scrum team follow Scrum practices and removes obstacles.
- **Developers –** Design, develop, test, and deliver the product increment.

### 4. What is a Sprint in Scrum?

- A **Sprint** is a fixed period in Scrum during which the team works to complete a selected set of tasks and create a usable product increment.
- **Typical duration:** 1 to 4 weeks, with 2 weeks being very common.

## 5. Jira Board Columns

| Column | Meaning |
|---|---|
| To Do | Tasks that are planned but work has not started yet. |
| In Progress | Tasks that the team is currently working on. |
| Done | Tasks that have been completed according to the project's definition of done. |
| Backlog | A list of pending tasks, requirements, bugs, and future work that has not yet been selected for active work. |

- **Simple flow:**

  Backlog → To Do → In Progress → Done

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## Session 5 — Git & GitHub

### 1. Git Commands

| Task | Git Command |
|---|---|
| Initialize a repository | git init |
| Stage all files | git add . |
| Commit with a message | git commit -m "Initial commit" |
| Push to remote | git push origin main |
| Pull latest changes | git pull origin main |

### 2. Difference between git fetch and git pull

- git fetch downloads the latest changes from the remote repository but does not merge them into your current branch.
- git pull downloads the latest changes and merges them into your current branch.

### 3. Create GitHub Repository and Push a Text File

You can do this using the following steps:

**Step 1: Create a repository on GitHub**

1. Open GitHub.
2. Click New repository.
3. Enter a repository name, for example: git-practice.
4. Click Create repository.

**Step 2: Create a text file locally**

- Create a file called:

  hello.txt

- Put some text inside it, for example:

  Hello, GitHub!

**Step 3: Run these commands in the folder**

```
git init

git add .

git commit -m "Add hello text file"

git branch -M main

git remote add origin https://github.com/YOUR_USERNAME/git-practice.git

git push -u origin main
```

- Replace YOUR_USERNAME with your GitHub username.
- **Repository link format:**

  https://github.com/YOUR_USERNAME/git-practice

- I can't create the repository in your GitHub account or provide your actual repo link without access to your account, so after you push it, submit your generated GitHub repository URL.

## 4. Fork vs Pull Request

- **Fork:** A fork creates your own copy of another person's GitHub repository in your GitHub account.
- **Pull Request (PR):** A pull request asks the original repository owner to review and merge your changes into their repository.

## 5. Create feature-test Branch, Commit, and Merge

**Step 1: Create and switch to the new branch**

```
git checkout -b feature-test
```

**Step 2: Make a change**

For example, create or modify feature.txt.

**Step 3: Stage and commit the change**

```
git add .
git commit -m "Add feature test"
```

**Step 4: Switch back to main**

```
git checkout main
```

**Step 5: Merge the branch**

```
git merge feature-test
```

Complete command flow

```
git checkout -b feature-test
git add .
git commit -m "Add feature test"
git checkout main
git merge feature-test
```

**Flow:**

main → feature-test → commit → main → merge feature-test

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